# GATE Computer Science & IT — 90-Day Execution Blueprint

Target: PSU Recruitment (AIR < 250 | Score: 780–880+) | Comprehensive Daily Roadmap

## 1. Performance & PSU Cutoff Benchmarks

| Parameter | Qualifying | Top IIT M.Tech | PSU Recruitment (IOCL/ONGC/NTPC) |
|---|---|---|---|
| Raw Marks (/100) | 28 – 32 | 60 – 68 | **72 – 82+ marks** |
| Normalized Score (/1000) | 350 – 420 | 700 – 760 | **780 – 880+** |
| All India Rank (AIR) | 10,000 – 15,000 | 500 – 1,200 | **AIR 1 to AIR 250** |

## 2. Daily Schedule Architecture (7.5 Hours/Day)

- **Block 1 (07:00 AM – 10:00 AM | 3.0 hrs):** Core Technical Concept Breakdown & Derivations.
- **Block 2 (11:30 AM – 02:00 PM | 2.5 hrs):** Rigorous Practice & PYQs (20–30 problems, timed solving).
- **Block 3 (05:00 PM – 07:00 PM | 2.0 hrs):** General Aptitude & Math (1.25 hrs) + Mistake Diary Review (0.75 hrs).

## 3. Phase 1: High-Yield Scoring Core (Days 1 to 30)

| Day | Core Technical Focus (5.5 hrs) | Evening Aptitude & Math (2.0 hrs) |
|---|---|---|
| Day 1 | C Data Types, Memory Layout, Operators, Pointer Arithmetic, Pointer to Arrays | Quantitative Aptitude: Ratios, Proportions, Percentages |
| Day 2 | Parameter Passing, Functions, Recursion Tracing, Stack Activation Records | Linear Algebra: Matrices, Determinants, Multiplications |
| Day 3 | 1D & 2D Arrays (Row/Column-major addressing), Dynamic Memory (malloc, free) | Quantitative Aptitude: Powers, Exponents, Logarithms |
| Day 4 | Singly, Doubly, Circular Linked Lists, Floyd's Cycle Detection | Linear Algebra: Rank, Row Echelon, System of Equations AX=B |
| Day 5 | Stacks: Implementations, Infix to Postfix/Prefix, Expression Evaluation | Verbal Aptitude: English Grammar & Verb-Noun Agreement |
| Day 6 | Queues: Circular Queue Boundary Checks, Deques, Stacks via Queues | Linear Algebra: Eigenvalues, Eigenvectors, Cayley-Hamilton |
| Day 7 | Consolidation Test: 30-Question Timed Test on C Programming & Linear DS | Analytical Aptitude: Series, Deductions, Analogy |
| Day 8 | Binary Trees: Properties, Traversals (Inorder, Preorder, Postorder) | Quantitative Aptitude: Permutations & Combinations |
| Day 9 | Binary Search Trees (BST): Insertion, Deletion cases, Successor/Predecessor | Calculus: Limits, Continuity, Differentiability |
| Day 10 | Binary Heaps: Min/Max Heap, O(n) Heapify derivation, Priority Queues | Spatial Aptitude: Transformation, Rotation, Mirroring |
| Day 11 | Asymptotic Analysis: Big-O, Omega, Theta, Master Theorem & Edge cases | Calculus: Maxima & Minima, Mean Value Theorem |
| Day 12 | Divide & Conquer: Merge Sort, Quick Sort (Pivots, Inversions) | Quantitative Aptitude: Elementary Statistics |
| Day 13 | Searching & Hashing: Collision handling (Chaining, Linear/Quadratic probing) | Calculus: Definite & Indefinite Integration |
| Day 14 | Consolidation Test: 30-Question Timed Test on Trees, Heaps, Sorting, Hashing | Analytical Aptitude: Syllogisms & Logic Deduction |
| Day 15 | Greedy: Fractional Knapsack, Huffman Coding, Activity Selection | Discrete Math: Propositional Logic, Truth Tables, CNF/DNF |
| Day 16 | Dynamic Programming 1: 0/1 Knapsack, Matrix Chain Multiplication, LCS | Discrete Math: First-Order Logic, Quantifiers, Scope |
| Day 17 | Dynamic Programming 2: Optimal BST, Subset Sum, Bellman-Ford, Floyd-Warshall | Discrete Math: Sets, Relations, Inclusion-Exclusion |
| Day 18 | Graph Traversals: BFS, DFS, Topological Sorting, Dijkstra's Algorithm | Discrete Math: Equivalence Relations & Classes |
| Day 19 | Minimum Spanning Trees: Prim's, Kruskal's, DSU, Cut & Cycle properties | Discrete Math: Posets, Hasse Diagrams, Lattices |
| Day 20 | Algebraic Structures: Semigroups, Monoids, Groups, Lagrange's Theorem | Verbal Aptitude: Reading Comprehension & Sequencing |
| Day 21 | Consolidation Test: Timed Test on Algorithms & Mathematical Logic | Spatial Aptitude: Paper Folding & 2D/3D Patterns |
| Day 22 | Combinatorics: Pigeonhole Principle, Permutations, Derangements | Quantitative Aptitude: Mensuration & Geometry |
| Day 23 | Recurrence Relations & Generating Functions: Solving Linear Recurrences | Verbal Aptitude: Vocabulary, Idioms & Phrases |
| Day 24 | Graph Theory 1: Degree Sequences, Handshaking Lemma, Connectivity | Quantitative Aptitude: Speed, Time, Distance, Work |

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| Day 25 | Graph Theory 2: Eulerian & Hamiltonian Graphs, Spanning Trees | Analytical Aptitude: Numerical Relations & Family Trees |
| Day 26 | Graph Theory 3: Planar Graphs (Euler's formula), Matching, Colouring | Quantitative Aptitude: Data Interpretation (Charts) |
| Day 27 | Discrete Mathematics Full PYQ Drill (GateOverflow 2-markers) | Aptitude Drill: 20 Mixed Questions |
| Day 28 | Comprehensive Revision of C, DS, Algorithms & Math Formulas | Mistake Diary Review & Correction |
| Day 29 | Full Subject Mock 1: Programming, DS & Algorithms (65 Marks, 90 mins) | Mock 1 In-depth Analysis & Error Remediation |
| Day 30 | Full Subject Mock 2: Discrete Mathematics & Engg Math (65 Marks, 90 mins) | Mock 2 In-depth Analysis & Error Remediation |

## 4. Phase 2: Systems, Theory & Databases (Days 31 to 60)

| Day | Core Technical Focus (5.5 hrs) | Evening Aptitude & Math (2.0 hrs) |
|---|---|---|
| Day 31 | Finite Automata: DFA Design, NFA to DFA Conversion, Epsilon-NFA | Probability: Sample Space, Conditional, Product Rule |
| Day 32 | Regular Expressions, DFA State Minimization (Myhill-Nerode) | Probability: Bayes' Theorem & Total Probability |
| Day 33 | Pumping Lemma for Regular Languages, Non-regularity proofs | Probability: Discrete Random Variables, Expectation |
| Day 34 | CFG & CFL: Derivations, Ambiguity, Pushdown Automata (DPDA/NPDA) | Probability: Continuous Variables, PDF, CDF |
| Day 35 | Pumping Lemma for CFLs, Non-CFL Identification, Chomsky Hierarchy | Probability: Uniform & Exponential Distributions |
| Day 36 | Turing Machines: Recursive/RE Languages, Undecidability, Rice's Theorem | Probability: Poisson, Binomial, Normal Distributions |
| Day 37 | Consolidation Day: TOC Closure & Decidability Master Matrix + Timed Test | General Aptitude: 15-Mark Full Sectional Test |
| Day 38 | Compiler Design: Lexical Analysis, Token generation, Lex regular rules | Quantitative Aptitude: Profit, Loss, Interest |
| Day 39 | Parsing 1: Top-down, First & Follow Sets, LL(1) Table & Conflicts | Spatial Aptitude: 2D/3D Rotation & Transformations |
| Day 40 | Parsing 2: Bottom-up, LR(0), SLR(1), CLR(1), LALR(1) Item Sets & Conflicts | Analytical Aptitude: Seating & Direction Sense |
| Day 41 | Syntax-Directed Translation (SDT), S/L-attributed, 3-Address Code | General Aptitude: Complex Tables Data Interpretation |
| Day 42 | Optimization: Basic Blocks, Flow Graphs, Liveness, Common Subexpressions | Linear Algebra & Probability: Mixed Revision Set |
| Day 43 | OS: System Calls, Fork Tracing, Threads, Scheduling (FCFS, SJF, RR) | Verbal Aptitude: Sentence Completion & Error Spotting |
| Day 44 | OS Synchronization: Critical Sections, Peterson's, Semaphores, Classics | Consolidation Test: Compiler & Synchronization Drill |
| Day 45 | Deadlocks: Necessary Conditions, RAG, Banker's Algorithm, Recovery | Engineering Math: Determinants & Limits Drill |
| Day 46 | Memory Management: Paging, Page Table Size, Multi-Level Paging Math | Quantitative Aptitude: Mixtures, Clocks, Calendars |
| Day 47 | Virtual Memory: TLB Hit/Miss (EMAT), Inverted Tables, Page Replacement | Spatial Aptitude: Mirror Imaging & Paper Cutting |
| Day 48 | File Systems & Disk Scheduling: i-nodes, SCAN, C-SCAN, LOOK, C-LOOK | Analytical Aptitude: Logical Deductions |
| Day 49 | DBMS: ER-Model, Cardinality, Relational Schema Mapping | Discrete Math: Logic & Lattices Quick Review |
| Day 50 | Relational Algebra, Tuple Calculus (TRC/DRC), Nested SQL Queries | Quantitative Aptitude: Permutations & Probability |
| Day 51 | Functional Dependencies, Keys, Normal Forms (1NF to BCNF), Lossless Joins | Consolidation Test: Operating Systems Timed Test |
| Day 52 | File Organization & Indexing: B & B+ Trees (Node Order, Record Pointers) | Engineering Math: Linear Equations & Bayes' Review |
| Day 53 | Transactions: ACID, Conflict vs View Serializability, Precedence Graphs | General Aptitude: 15-Mark Full Sectional Test |
| Day 54 | Concurrency Control: 2PL, Strict 2PL, Timestamp Ordering Protocol | Analytical Aptitude: Numerical Reasoning Drills |

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| Day 55 | Digital Logic 1: Boolean Algebra, K-Maps, Tabular Method (Quine-McCluskey) | Calculus & Probability: Formula Review Set |
| Day 56 | Digital Logic 2: Number Representations, 2's Complement, IEEE 754 Floating | Spatial Aptitude: 3D Assembling from 2D Nets |
| Day 57 | Combinational Circuits: Adders, MUX, Decoders, Circuit Implementations | Quantitative Aptitude: Advanced DI Test |
| Day 58 | Sequential Circuits: Flip-Flops (JK, D, T), Counters (Mod-N), State Diagrams | Consolidation Test: DBMS & Digital Logic Test |
| Day 59 | Full Subject Mock 3: Operating Systems & Compiler Design (65 Marks) | Mock 3 In-depth Analysis & Error Remediation |
| Day 60 | Full Subject Mock 4: DBMS & Theory of Computation (65 Marks) | Mock 4 In-depth Analysis & Error Remediation |

## 5. Phase 3: Architecture, Networks & Full Mocks (Days 61 to 90)

| Day | Core Technical Focus (5.5 hrs) | Evening Aptitude & Math (2.0 hrs) |
|---|---|---|
| Day 61 | COA 1: Machine Instructions, Addressing Modes, Hardwired/Microprogrammed CU | Verbal Aptitude: Reading Comprehension |
| Day 62 | COA 2: Memory Hierarchy, Cache Mapping (Direct, Set-Assoc), AMAT Math | Quantitative Aptitude: Probability Drills |
| Day 63 | COA 3: Instruction Pipelining, Structural/Data/Control Hazards, Speedup | Engineering Math: Matrices & Calculus |
| Day 64 | COA 4: I/O Interface: Programmed, Interrupt-Driven, DMA Cycle Stealing | Spatial Aptitude: Mixed Spatial Paper Test |
| Day 65 | Networks 1: Switching, Framing, Delays (Tx, Prop), CRC & Checksums | Analytical Aptitude: Syllogisms & Logic |
| Day 66 | Networks 2: Flow Control (Stop & Wait, GBN, SR), CSMA/CD, Ethernet | General Aptitude: 15-Mark Full Sectional |
| Day 67 | Networks 3: IPv4 Header, Fragmentation Offsets, CIDR, Subnetting, Routing | Consolidation Test: COA & Networks Timed Drill |
| Day 68 | Networks 4: TCP Congestion Control (Slow Start, Congestion Avoidance, AIMD) | Quantitative Aptitude: Full Speed Drill |
| Day 69 | Networks 5: Application Layer: DNS, HTTP Persistent/Non-persistent RTT | Discrete Math: Monoids & Graphs Quick Review |
| Day 70 | Full Subject Mock 5: COA & Digital Logic (65 Marks, 90 mins) | Mock 5 Detailed Analysis & Error Log Scrub |
| Day 71 | Full Subject Mock 6: Networks & General Aptitude (65 Marks, 90 mins) | Mock 6 Detailed Analysis & Error Log Scrub |
| Day 72 | Multi-Subject Test 1: Programming + DS + Algorithms + OS (50 Marks) | Multi-Subject 1 Error Review |
| Day 73 | Multi-Subject Test 2: TOC + Compiler + DBMS + Networks (50 Marks) | Multi-Subject 2 Error Review |
| Day 74 | Multi-Subject Test 3: Discrete Math + Engg Math + Digital + COA (50 Marks) | Master Formula Sheet Consolidation |
| Day 75 | Full-Length Mock 1 (FLT 1) — 65 Questions / 100 Marks (09:30 - 12:30) | Full Root-Cause Analysis (3.5 hrs) |
| Day 76 | Weakness Remediation: Revision of bottom 2 subjects from FLT 1 | Aptitude & Math Timed Speed Drill |
| Day 77 | Full-Length Mock 2 (FLT 2) — 65 Questions / 100 Marks (09:30 - 12:30) | Full Root-Cause Analysis & Log Update |
| Day 78 | Master Formula Review (Paging, Caching, Pipelining, Master Theorem) | Solve 15 NATs on GateOverflow with timer |
| Day 79 | Full-Length Mock 3 (FLT 3) — 65 Questions / 100 Marks (09:30 - 12:30) | Full Root-Cause Analysis (3.5 hrs) |
| Day 80 | TOC Closure Table, Decidability, and Compiler Parsing Tables Revision | Spatial & Analytical Aptitude Review |
| Day 81 | Full-Length Mock 4 (FLT 4) — 65 Questions / 100 Marks (09:30 - 12:30) | Full Root-Cause Analysis (3.5 hrs) |
| Day 82 | Operating Systems Concurrency, Deadlocks & B+ Tree Indexing Review | Engineering Math High-Yield Formulas |
| Day 83 | Full-Length Mock 5 (FLT 5) — 65 Questions / 100 Marks (09:30 - 12:30) | Full Root-Cause Analysis (3.5 hrs) |
| Day 84 | NAT-Only Speed Drill: 40 Numerical Answer Type questions timed | Verbal & Quantitative Final Polish |

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| Day 85 | Full-Length Mock 6 (FLT 6) — 65 Questions / 100 Marks (09:30 - 12:30) | Full Root-Cause Analysis (3.5 hrs) |
| Day 86 | Complete Mistake Diary Scrub: Re-solve every logged error | Re-attempt all negative marks from FLTs |
| Day 87 | Full-Length Mock 7 (FLT 7 — Final Simulation) — 65 Qs / 100 Marks | Final Strategy & Question Skip Check |
| Day 88 | Final Comprehensive Formula & Identity Review across all 10 subjects | Zero new problems; consolidate memory |
| Day 89 | Light Mistake Diary Review & Virtual Calculator Shortcuts Refresh | Mental Relaxation & Sleep Optimization |
| Day 90 | Exam Logistics: Verify Admit Card, Original Photo ID, Center Route | Rest & High Performance Priming |

## 6. Authoritative Textbook References & Strategy

- **Algorithms & DS:** Cormen, Leiserson, Rivest, Stein (CLRS) — Focus on Chapters 1–4, 11, 12, 15, 16, 22–24.
- **Operating Systems:** Silberschatz, Galvin, Gagne — Focus on Processes, Synchronization, Deadlocks, Memory Management.
- **Databases:** Silberschatz, Korth, Sudarshan — Relational Algebra, Normalization, B+ Trees, Transactions.
- **Theory of Computation:** Peter Linz — Automata, Regular Languages, CFG, Turing Machines, Undecidability.
- **Computer Networks:** Kurose & Ross — Framing, CRC, Sliding Window, Subnetting, TCP Congestion Control.
- **Computer Organization:** Carl Hamacher — Cache Mapping, Pipeline Hazards & Throughput, Addressing Modes.
- **Discrete Mathematics:** Kenneth H. Rosen — Propositional Logic, Equivalence Relations, Graph Invariants.
- **PYQ Resource:** GateOverflow PDF Volumes 1 & 2 (Official past papers 1987–2026). Prioritize 2012–2026.

## 7. Negative Marking & Exam-Room Protocols

- **Three-Pass Solving Technique:** Pass 1 (0–60m): Direct 1-markers and trivial 2-markers (aim for 30–35 marks). Pass 2 (60–140m): Multi-step NATs and code traces. Pass 3 (140–180m): Verification and flagged questions.
- **Zero Guessing on MCQs:** A single 2-mark incorrect MCQ incurs a -0.66 deduction (effective loss of 2.66 marks). 3 wild guesses can forfeit ~8 marks, dropping rank from Top 250 to outside 1,500.
- **NAT Advantage:** Numerical Answer Type questions carry NO negative marking. If your answer is within a verified mathematical bound, never leave it blank.
- **Virtual Calculator:** Never compute by hand for exponents or logarithms. Practice exclusively with the official TCS on-screen calculator.